

KAHRAMANMARAS, Turkey

WMO: 172550

Lat: 37.600N

Lon: 36.933E

Elev: 1877

StdP: 13.73

Time Zone: 3.00 (E03)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.3	30.5	8.2	9.0	40.2	13.6	11.7	43.8	20.8	44.1	18.7	43.9	3.6	80	0.435

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	24.5	102.3	69.2	100.0	68.5	97.0	68.9	78.0	90.6	76.0	88.6	74.3	87.4	6.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.5	138.6	85.4	72.5	129.1	84.0	70.2	119.3	82.9	43.2	90.8	41.1	88.2	39.3	87.3	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.3	21.7	19.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	64.1	41.6	45.6	53.5	61.3	68.7	78.5	85.0	85.5	79.1	67.4	55.1	46.0	(6)
(7)		DBStd	15.90	5.10	5.69	5.62	5.76	5.31	4.13	3.31	3.16	4.64	5.57	5.91	4.74	(7)
(8)		HDD50	596	263	144	29	1	0	0	0	0	0	25	134	(8)	
(9)		HDD65	2715	727	542	356	140	25	0	0	1	39	297	588	(9)	
(10)		CDD50	5724	1	21	139	340	580	856	1085	1101	872	539	179	11	(10)
(11)		CDD65	2368	0	0	1	29	140	406	620	636	422	114	0	0	(11)
(12)		CDH74	27981	0	1	9	340	1385	4333	8118	8227	4588	970	10	0	(12)
(13)		CDH80	14357	0	0	1	74	478	2068	4597	4585	2278	276	0	0	(13)
(14)	Wind	WSAvg	5.6	3.2	3.1	4.4	4.4	5.4	9.3	10.7	10.3	7.3	3.6	2.3	3.5	(14)
(15)	Precipitation	PrecAvg	27.3	4.6	4.3	3.5	2.8	1.8	0.3	0.1	0.1	0.6	1.8	3.2	4.6	(15)
(16)		PrecMax	42.6	12.3	8.2	9.8	6.6	5.6	1.1	0.3	0.2	3.7	5.2	9.4	11.3	(16)
(17)		PrecMin	17.0	0.6	0.8	0.4	0.7	0.4	0.0	0.0	0.0	0.1	0.0	0.0	(17)	
(18)		PrecStd	5.7	2.9	2.1	2.0	1.5	1.1	0.3	0.1	0.1	0.7	1.2	2.2	2.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.4	69.0	74.7	86.5	94.4	100.9	106.2	105.8	102.1	90.0	75.4	64.7	(19)
(20)		MCWB	46.2	51.7	56.6	60.3	64.5	68.0	70.2	68.8	64.7	64.2	55.3	51.0	(20)	
(21)		2%	DB	54.0	62.2	70.3	81.3	89.4	97.0	102.6	102.2	98.4	86.3	72.1	59.5	(21)
(22)		MCWB	44.3	49.6	53.5	58.4	62.6	67.2	69.2	70.7	64.3	62.8	54.0	47.4	(22)	
(23)		5%	DB	51.7	58.8	68.0	77.4	85.1	93.5	100.1	99.8	94.8	82.7	69.4	56.8	(23)
(24)		MCWB	43.4	47.9	52.5	57.7	61.9	66.4	68.3	70.4	64.8	60.9	53.7	46.6	(24)	
(25)		10%	DB	49.6	55.4	64.7	74.2	81.8	90.4	97.0	96.8	91.3	79.2	66.2	53.6	(25)
(26)		MCWB	43.2	46.1	51.3	56.7	60.5	66.2	69.5	70.6	65.7	59.4	52.2	46.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.5	53.3	59.0	64.7	68.1	75.0	80.0	80.2	75.5	71.3	60.0	53.9	(27)
(28)		MCDWB	53.5	65.5	70.6	82.2	84.5	88.1	93.5	93.4	85.0	78.6	69.4	59.3	(28)	
(29)		2%	WB	47.0	50.9	56.2	61.6	65.9	73.0	78.2	78.3	73.2	67.4	56.9	50.7	(29)
(30)		MCDWB	51.3	60.0	66.5	75.2	81.0	88.6	91.2	90.8	84.1	79.2	66.9	56.6	(30)	
(31)		5%	WB	45.6	49.1	54.4	59.6	64.3	71.1	76.2	76.5	71.6	64.3	55.3	49.2	(31)
(32)		MCDWB	49.6	56.8	63.9	72.0	79.7	86.4	89.1	89.1	83.2	76.5	64.5	53.5	(32)	
(33)		10%	WB	44.1	47.5	52.7	58.1	62.8	69.4	74.3	74.7	69.9	62.5	53.8	47.9	(33)
(34)		MCDWB	47.8	53.7	61.7	70.4	78.0	83.9	87.8	87.8	82.2	74.7	63.1	51.8	(34)	
(35)	Mean Daily Temperature Range		MDBR	12.3	15.3	18.2	20.9	22.1	22.8	25.2	24.5	25.0	22.2	20.2	13.8	(35)
(36)		5% DB	MCDBR	17.7	20.8	23.5	27.1	28.7	27.8	30.6	29.1	30.5	26.7	25.2	20.8	(36)
(37)			MCWBR	9.8	10.3	10.3	10.3	9.6	8.8	10.2	8.8	10.4	10.2	11.5	11.5	(37)
(38)			MCDBR	12.7	17.7	19.9	23.2	25.5	24.1	25.2	24.7	24.6	22.9	21.7	15.7	(38)
(39)		5% WB	MCWBR	7.9	9.0	9.4	9.6	9.4	8.6	8.5	9.2	9.4	9.6	10.8	9.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.341	0.367	0.406	0.452	0.430	0.383	0.377	0.392	0.363	0.392	0.353	0.335	(40)	
(41)		taud	2.295	2.225	2.151	2.055	2.181	2.343	2.368	2.308	2.387	2.270	2.340	2.358	(41)	
(42)		Ebn at Noon	262	270	270	263	271	284	285	278	281	258	256	257	(42)	
(43)		Edh at Noon	31	37	44	52	47	40	38	40	35	36	30	27	(43)	
(44)	All-Sky Solar Radiation	RadAvg	676	941	1343	1700	2069	2469	2498	2216	1817	1252	875	641	(44)	
(45)		RadStd	110	155	116	161	119	114	79	77	78	93	93	115	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page